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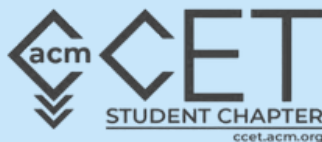
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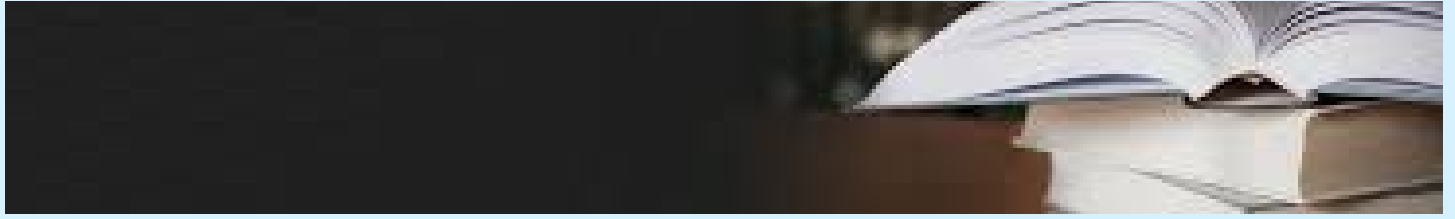
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PUBLICATIONS



1. Journal Articles

A Novel Approach for Evaluating Spatial-Temporal Synergy in Hybrid CNN-RNN and Vision Transformer Architectures

Authors - Viren Passi, Sudhakar Kumar, Sunil K Singh, Shreya Verma, Varsha Arya, Valerie Tang, Brij B Gupta, Kwok Tai Chui

Optimized CNN-based ensemble deep learning approach for potato leaf disease detection with data augmentation

Authors - Achin Jain, Arun Kumar Dubey, Sunil K Singh, Arvind Panwar, Neha Gupta, Sudhakar Kumar, Varsha Arya, Wadee Alhalabi, Shin-Hung Pan, Bassma Saleh Alsulami, Brij B Gupta

On-Chip Intelligence

Authors - Sunil K Singh, Varsha Arya

2. Conference Papers

A Deep Learning-based Static Gesture Categorization and Interpretation of Indian Sign Language (SGCIISLang) Model Using Optimized Gated Recurrent Unit (GRU)

Authors - Animesh Singh, Sunil K Singh, Ajay Mittal, Mayank Sharma, Preeti Aggarwal

Improving predictive reliability and automation of smart grids using the StarNet ensemble model

Authors - Amit Chhabra, Sunil K Singh, Sudhakar Kumar, Manraj Singh, Utkarsh Chauhan, Saksham Arora, Wadee Alhalabi, Varsha Arya, Bassma Saleh Alsulami, Ching-Hsien Hsu, Brij B Gupta



Comparative Analysis of Random Forest, Support Vector Regression, Gradient Boosting and Phyllo-ViT Model for the Prediction of Air Pollutants

Authors - Neelam Yadav, Sunil K Singh, Dinesh Sharma

AI-Driven Deep Learning for Real-Time DDoS Detection in Software-Defined Networks

Authors - Sudhakar Kumar, Sunil K Singh, Rakesh Kumar, Satyam Kumar Sainy, Anand Kumar Yadav, Shabeg Singh Gill

Ensemble-Driven Malware Detection with Anomaly Override Capabilities

Authors - Viren Passi, Sudhakar Kumar, Sunil K Singh, Shivam Jindal, Prince Raj, Varsha Arya, Kwok Tai Chui, Brij B Gupta

3. Book Chapters

IoT-XPLAIN: Intelligent Fault Detection and Natural Language Explanation for IoT Systems Using ML and LLMs

Authors - Harshit Vashisht, Sunil K Singh, Sudhakar Kumar, Nipun Singh, Varsha Arya, Kwok Tai Chui, Brij B Gupta

EVENTS



The Commission for Scientific and Technical Terminology (CSTT), Ministry of Education, organized an Expert Advisory Committee meeting hosted by the School of Punjabi Studies at Panjab University, Chandigarh, to prepare a trilingual (English-Hindi-Punjabi) glossary for Artificial Intelligence and Machine Learning. Inaugurated by MP Satnam Singh Sandhu, Vice Chancellor Prof. Renu Vig, and CSTT Assistant Director Deepak Kumar, the five-day workshop focuses on standardizing technical terminology in Indian languages in direct alignment with the National Education Policy (NEP) 2020.

- Sh. Satnam Singh Sandhu (Rajya Sabha MP) highlighted the initiative's alignment with the National Education Policy (NEP) 2020, emphasizing that translating technical concepts into regional languages bridges the digital divide and makes advanced technologies accessible to all students.
- Prof. Renu Vig (Vice Chancellor, PU) stressed that standardized local terminology is vital for building inclusive digital ecosystems and fostering research innovation.
- Sh. Deepak Kumar (CSTT) outlined the Ministry's ongoing commitment to standardizing technical terminology in Indian languages to facilitate regional technical education.

In conclusion, this joint initiative by CSTT and Panjab University represents a significant step toward making advanced technical education inclusive and accessible. By bridging language barriers in cutting-edge fields like Artificial Intelligence and Machine Learning, the trilingual glossary not only fulfills the core vision of NEP 2020 but also empowers regional students to meaningfully engage with the future of global technology.

Key Insights



Deep Learning for Early Identification of Neurodegenerative Diseases

ARTICLE

Abstract

Degenerative neurological conditions including Alzheimer's, Parkinson's, and Amyotrophic Lateral Sclerosis are diseases characterized by the progressive degeneration of neurons several years before any clinical appearance. One of the biggest challenges in modern medicine is the early identification of these diseases. Deep learning is one field of artificial intelligence that has been able to demonstrate remarkable results in identifying patterns in biomedical sciences, thus facilitating early diagnosis [1], [5]. This article presents an overview of how deep learning techniques have been utilized in diagnosing neurodegenerative diseases at their early stages.

Introduction

Neurodegenerative diseases (NDs) are progressive illnesses caused by the degeneration of neurons in the CNS. Often, these NDs are clinically silent at the initial stage of their development, thereby reducing the efficacy of treatment. An early diagnosis is crucial since it makes it possible to recognize the physiological mechanisms of a disease occurring 10–15 years before the manifestation of symptoms [1], [6]. The advent of deep learning technologies in medicine allowed for automatic processing of large-scale datasets. Unlike conventional approaches based on hand-crafted features, deep learning systems can learn data representation independently [5], [8].

2. Deep Learning in Neurodegenerative Disease Detection

The deep learning (DL) models utilize artificial neural networks that use multiple layers to represent the hierarchical nature of data [6]. DL is more advanced than conventional machine learning because the former does not need any feature extraction process to enhance accuracy of predictions [5], [6]. Examples of areas in which DL-based techniques have demonstrated effectiveness include the following:

Identification and prediction of diseases [6]

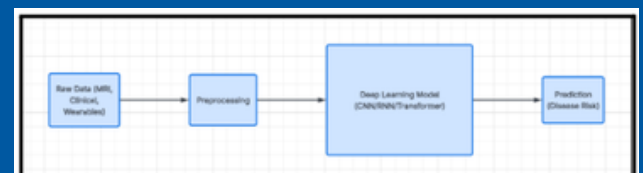
Clinical image interpretation [5], [8]

Detection of biomarkers [4]

Disease modeling [3]

Deep learning uses neural networks to automatically learn patterns from complex medical data. It follows a structured pipeline starting from data collection to final prediction is shown in fig 1.

Fig. 1: Deep Learning pipeline for early detection of neurodegenerative diseases.



4. Models Used

Deep learning algorithms are essential for analysis of various kinds of medical datasets for early diagnosis of neurodegenerative diseases. Among the most popular models, one can mention Convolutional Neural Networks (CNN). This algorithm is very commonly used for image datasets such as MRI scans or PET. CNN uses numerous filters in order to extract crucial spatial features such as edges, texture, and other structural changes in a brain. Thanks to this, early diagnosis of brain atrophy can be made [5], [7]. The next set of models worth considering includes recurrent neural networks (RNN), as well as advanced versions of the latter such as LSTM networks. This model is able to preserve the information about the previous state and is therefore able to analyze temporal data including patient history, cognition scores, and repeat measures in order to evaluate the disease progression [3], [6]. More advanced and complex models to be mentioned in this context are transformer models. Thanks to the use of attention mechanisms, this algorithm is able to focus on the most relevant information and, consequently, to enhance predictions for larger and multimodal datasets including imaging, clinical, and genomics data [3], [6]. Convolutional Neural Networks (CNNs) are widely used for analyzing medical images such as MRI and PET scans. They extract important spatial features like edges and textures to detect brain abnormalities. Deep learning models such as CNN, RNN, and transformers are widely used in medical applications. Their functions are summarized in Table 1.

5. Applications

Deep learning approaches have been successfully employed in disease identification across several types of medical and behavioral datasets. For instance, during the early diagnosis of Alzheimer's disease, CNNs can be used to assess MRI and PET scans to detect

the initial development of brain changes, including amyloid plaque buildup and hippocampal atrophy [2], [7]. The occurrence of these physiological changes usually takes place years ahead of the memory impairments, making it possible to diagnose and intervene in the disease. For Parkinson's disease, deep learning algorithms are utilized to evaluate motion-related characteristics like tremors, voice variations, gait disturbances, and penmanship [4], [6]. Patients can thus be evaluated through non-invasive means such as wearables and voice records. Moreover, deep learning can be employed in identifying early signs and tracking the development of amyotrophic lateral sclerosis (ALS). Using clinical records and neurological markers, these deep learning models can determine motor neuron degeneration and ALS progression rate [3], [6].

6. Challenges

to be addressed when using deep learning for neurodegenerative disease detection. Firstly, there is a shortage of high-quality medical data with proper labeling for use during training. Secondly, deep learning systems are known for being black boxes, which poses a problem of transparency that makes it hard for medical professionals to trust the decision-making process. Thirdly, the performance of models might differ depending on the dataset used because of the differences in how medical data was collected. Medical data is highly sensitive, and ensuring privacy and security is a major challenge in deploying deep learning systems in healthcare [6], [7].

7. Conclusion

The use of deep learning has proven to be effective in the early detection of neurological disorders through the analysis of complicated medical data, which can easily be detected through patterns that would have been hard to detect through conventional approaches. While some problems like data scarcity and non-interpretability continue to hinder the use of deep learning, continued technological advancements will help solve these problems. In the future, deep learning will prove to be very important in the early diagnosis of diseases and personalizing treatments.

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WRITTEN BY - TANVI (2024-2028)

Light Speed: The Strategic Ascent of Photonic Integrated Circuits

ARTICLE

Abstract

Traditional electron-based computing is approaching a saturation point as physical and thermal limitations increasingly constrain performance scaling. As Moore's Law slows, challenges such as heat dissipation, interconnect latency, and bandwidth bottlenecks have become dominant barriers to further progress. Photonic Integrated Circuits (PICs), which utilize light instead of electrical signals for data transmission, present a compelling alternative. By leveraging the inherent advantages of photons—such as high speed, low loss, and parallelism—PICs enable significantly higher bandwidth and improved energy efficiency, making them particularly well-suited for modern artificial intelligence (AI) workloads [1][2].

This paper explores the architectural advantages of silicon photonics, including bandwidth density, reduced energy consumption, and resilience to interference. It also examines the major challenges hindering widespread adoption, such as manufacturing complexity and the lack of standardization across the industry [2][5]. Furthermore, it evaluates the evolving AI-driven landscape in 2026, where photonic interconnects are rapidly transitioning from experimental technologies to essential components in high-performance computing systems [4][6].

INTRODUCTION

For decades, the computing industry has relied on transistor scaling to drive performance improvements. Increasing transistor density enabled faster processors and more powerful systems, forming the foundation of exponential growth in computational capabilities. However, while computational units have advanced significantly, the communication infrastructure connecting them has not kept pace. This imbalance has resulted in what is widely referred to as the interconnect bottleneck, a critical limitation that restricts overall system performance. Prior research indicates that as computational systems scale, inefficiencies in data movement and parallel execution increasingly dominate performance and energy costs [9][10].

In modern computing environments, particularly in data centers and AI clusters, the need for efficient data movement has become as important as computation itself. Large-scale AI models require massive parallel processing across distributed GPU clusters, where thousands of processing units must exchange data continuously. Traditional copper-based interconnects struggle in this context due to signal degradation, electromagnetic interference, and increasing power consumption over longer distances [7]. These inefficiencies not only limit performance but also significantly increase operational costs in large-scale deployments. In highly parallel intelligent systems,

managing communication overhead and synchronization becomes a critical challenge, further emphasizing the need for more efficient interconnect architectures [9][10].

ARCHITECTURAL BENEFITS

This makes photonic interconnects inherently more energy-efficient, especially in scenarios involving high data throughput. Additionally, reduced heat generation simplifies thermal management, enabling more compact and scalable system designs. Efficient data movement is also a key factor in minimizing speculative and parallel execution overheads in modern intelligent systems, further highlighting the importance of energy-efficient interconnect technologies [2][3][9][10].

Signal Integrity and Reliability

Another key advantage of photonic systems is their immunity to electromagnetic interference (EMI). Electrical signals are highly susceptible to noise and crosstalk, particularly in densely packed circuits operating at high frequencies. This can lead to data corruption and reduced reliability, requiring complex error correction mechanisms..

Latency and Scalability Considerations

In addition to bandwidth and energy efficiency, photonic interconnects offer advantages in latency and scalability. Optical communication can achieve lower latency over long distances compared to electrical signaling, particularly in distributed computing environments. As systems scale across racks, clusters, or even data centers, the performance benefits of photonic interconnects become increasingly pronounced. This is particularly relevant in large-scale parallel systems, where reducing communication latency directly impacts overall system efficiency and scalability [9][10].

moreover, the ability to integrate photonic components with existing silicon-based manufacturing processes enables gradual adoption without requiring a complete overhaul of current infrastructure. This compatibility positions silicon photonics as a practical and scalable solution for future computing systems. Bandwidth Density and Parallelism

One of the most significant advantages of photonic systems lies in their ability to achieve extremely high bandwidth density. This is primarily enabled by Wavelength Division Multiplexing (WDM), a technique that allows multiple data streams to be transmitted simultaneously over a single optical channel by using different wavelengths of light.

This capability dramatically increases throughput without increasing physical complexity. Such parallelism at the communication level complements advances in hardware-supported parallel architectures designed to reduce computational bottlenecks in intelligent systems [9][10]. In large-scale systems such as hyperscale data centers, this translates to more efficient utilization of interconnect infrastructure and reduced congestion in communication pathways [7].

Energy Efficiency and Thermal Advantages
Energy efficiency has become a critical concern in modern computing, particularly as data centers scale to meet growing demand. Electrical interconnects suffer from resistive losses, where energy is dissipated as heat during signal transmission. This not only reduces efficiency but also necessitates additional cooling mechanisms, further increasing power consumption.

In contrast, optical signals experience significantly lower attenuation, allowing them to travel longer distances with minimal energy loss.

CHALLENGES AND LIMITATIONS

- **Manufacturing Complexity and Material Constraints**

Despite their advantages, Photonic Integrated Circuits face significant challenges in manufacturing and scalability. While silicon photonics benefits from compatibility with established CMOS fabrication processes, one of the most critical limitations lies in efficient on-chip light generation. Silicon, by its physical nature, is an indirect bandgap material and therefore an inefficient emitter of light.

To address this, manufacturers must integrate external materials such as indium phosphide (InP) or other III-V semiconductors, which are capable of efficient light emission. However, this hybrid integration introduces substantial complexity into the fabrication process. Aligning different material systems with high precision increases production costs and can negatively impact yield, making large-scale manufacturing more difficult [2][3].

- **Design Ecosystem and Lack of Standardization**

Another major barrier to widespread adoption is the relative immaturity of the photonics design ecosystem. Unlike traditional electronic design, which benefits from decades of standardization, robust toolchains, and well-established fabrication workflows, photonic design is still evolving.

Currently, there is a lack of standardized design libraries, simulation tools, packaging techniques, and testing methodologies. This forces companies and research institutions to develop custom solutions tailored to their specific implementations. Such fragmentation slows innovation and increases development costs, as interoperability between different systems becomes limited [5][8].

Efforts by industry organizations to establish common standards are underway, but widespread adoption will require coordinated collaboration across multiple stakeholders, including semiconductor manufacturers, system designers, and software tool providers.

6. Challenges

There are numerous problems that need to be addressed when using deep learning for neurodegenerative disease detection. Firstly, there is a shortage of high-quality medical data with proper labeling for use during training. Secondly, deep learning systems are known for being black boxes, which poses a problem of transparency that makes it hard for medical professionals to trust the decision-making process. Thirdly, the performance of models might differ depending on the dataset used because of the differences in how medical data was collected. Medical data is highly sensitive, and ensuring privacy and security is a major challenge in deploying deep learning systems in healthcare [6], [7].

MARKET LANDSCAPE IN 2026

Growth Driven by AI and Data Centers
By 2026, silicon photonics has reached a critical inflection point. The global market is estimated to be valued at approximately \$3.5 billion, with rapid growth driven primarily by the increasing demand for high-performance computing and artificial intelligence applications [2][3]. AI workloads, particularly those involving large language models and generative systems, require immense data movement between processing units.

- Traditional interconnect technologies are increasingly unable to meet these bandwidth demands efficiently, creating a strong incentive for the adoption of photonic solutions.
- Hyperscale data centers, operated by major technology companies, are at the forefront of this transition. These facilities prioritize performance-per-watt and scalability, both of which align closely with the strengths of photonic interconnects.
- Emergence of Co-Packaged Optics (CPO)
- One of the most significant technological developments in this domain is Co-Packaged Optics (CPO). In this approach, photonic components are integrated directly alongside electronic processors within the same package, rather than being connected via external optical modules.
- This integration reduces the distance that electrical signals must travel before being converted into optical signals, thereby minimizing energy loss and latency. As a result, CPO offers substantial improvements in both performance and power efficiency compared to traditional pluggable optical transceivers [5][8].
- CPO is increasingly being viewed as a key enabler for next-generation data center architectures, particularly as bandwidth requirements continue to scale exponentially.
- Role of AI in Accelerating Photonics Adoption
- Artificial intelligence is not only a driver of demand but also a contributor to innovation in photonics. Research into photonic neural networks suggests that optical systems could be used to accelerate certain types of computations, particularly those involving matrix operations and parallel data processing [6].
- While still in early stages, such approaches indicate the potential for photonics to move beyond interconnects and play a direct role in computation itself. Efficient handling of parallel workloads and reduction of speculative execution overheads are critical in such environments, further motivating the adoption of high-bandwidth, low-latency photonic interconnects [4][6][9][10].

FUTURE DIRECTIONS AND EMERGING TRENDS

expected to shape the evolution of photonic integrated circuits:

- Improved Light Sources: Advances in heterogeneous integration techniques may enable more efficient and cost-effective on-chip lasers.
- Standardization Efforts: Industry-wide initiatives are likely to establish common frameworks for design, packaging, and testing, reducing fragmentation.
- Edge Integration: As photonic components become more compact and efficient, their deployment may extend beyond data centers to edge devices and specialized hardware.
- Photonic Computing: Continued research into photonic neural networks and optical computing could unlock entirely new paradigms of computation.

These developments will determine how quickly photonics transitions from a complementary technology to a foundational element of computing systems.

CONCLUSION

Photonic Integrated Circuits represent a transformative shift in the design and architecture of modern computing systems. As traditional electronic approaches encounter fundamental physical limitations, photonics offers a viable path forward by leveraging the unique properties of light for data transmission. When combined with advances in parallel computing

- architectures and optimization techniques, photonic systems can significantly improve overall system efficiency and scalability [9] [10].
- With advantages such as high bandwidth density, superior energy efficiency, and immunity to electromagnetic interference, PICs are particularly well-suited for the demands of AI-driven workloads and large-scale data center operations [1][2]. At the same time, challenges related to manufacturing complexity and lack of standardization continue to hinder widespread adoption.
- The progress of silicon photonics over the next decade will depend on overcoming these barriers through innovation in materials, fabrication techniques, and ecosystem development. If successful, photonic technologies have the potential to extend beyond data centers and become integral to edge computing, communication systems, and even computational architectures themselves.
- In this context, photonic integrated circuits are not merely an incremental improvement but a foundational advancement that could redefine the future of computing.

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**WRITTEN BY - HITESH SAINI
(2023-2027)**

Why Deep Quantum Circuits Forget Their Own History - And Why That's a Crisis

ARTICLE

Abstract

Variation quantum algorithms (VQAs) are widely seen as the most viable path toward quantum advantage, yet they face a fundamental scalability crisis known as the barren plateau. As quantum circuits grow in depth to handle complex tasks, they increasingly suffer from noise-induced information loss, effectively "forgetting" their historical parameter settings. This article explores how hardware noise, architectural connectivity, and algebraic structures converge to flatten the optimization landscape. By examining the roles of contractive maps, the linear depth barrier, and the curse of dimensionality, we highlight why deep circuits lose their training signal and how engineered strategies like controlled dissipation are being used to restore circuit memory and algorithmic viability.

INTRODUCTION

The Noisy Intermediate-Scale Quantum (NISQ) era is defined by devices that possess significant qubit counts but remain limited by high error rates. Variation quantum circuits serve as a bridge in this landscape, employing classical optimizers to train quantum systems for machine learning and chemistry. However, as these circuits become deeper, they encounter "noise-induced barren plateaus" (NIBPs), where the gradient of the cost function disappears exponentially. This loss of sensitivity to trainable parameters creates a flat optimization landscape where finding a solution becomes statistically impossible.

The Contractive Nature of Noise Maps

The primary reason deep circuits lose their history is the mathematical property of noise channels known as contractivity. In a layered error model, parameterized unitaries are alternated with completely positive trace-preserving (CPTP) noise maps. These maps naturally make quantum states more indistinguishable over time, a process quantified by the contraction coefficient or Lipschitz constant of the map. For strictly contractive noise, such as amplitude damping or certain Pauli channels, repeated applications drive the system toward a unique fixed point regardless of the initial parameters chosen. Consequently, as the total number of layers in a circuit increases, the system exponentially forgets its dependency on historical parameters, rendering further depth useless for training. (Schumann et al., 2024)

Hardware Architecture as a Scaffold for Noise propagation

The loss of information is not uniform across all devices; rather, it is heavily influenced by the connectivity architecture of the processor. Quantum architectures can be represented as connected graphs where nodes are qubits and edges represent two-qubit gate pathways (Goyal et al., 2024). Research using machine learning shows that superconducting systems are highly sensitive to global connectivity measures like betweenness centrality and spectral entropy.

In contrast, semiconductor quantum dot platforms are more sensitive to the overall system scale and node count. These topological features act as identifiable scaffolds that determine how environmental noise propagates through the system, either accelerating or slowing the rate at which the circuit loses its coherence and historical data. (Fu et al., 2026)

The Taxonomy of Scalability Barriers

Barren plateaus represent a multifaceted crisis in quantum computing, categorized into different types based on their origin. Standard Barren Plateaus (SBP) occur when highly expressive circuits approximate a unitary 2-design, causing the gradient variance to vanish exponentially with the number of qubits. Beyond expressivity, factors such as the locality of measurements and the level of entanglement in the initial state also contribute to this flattening. Mitigation strategies range from specialized parameter initialization—like using identity blocks or Gaussian distributions—to incremental layer-wise training. Despite these efforts, the fundamental issue remains: as the model size increases, the optimization landscape becomes a featureless "plateau" where most gradient-based methods fail. (Cunningham & Zhuang, 2025)

The Linear Depth Barrier and Pauli Noise

A critical analytical result shows that under local Pauli noise, the training signal vanishes exponentially if the circuit depth grows linearly with the number of qubits. This phenomenon, known specifically as a Noise-Induced Barren Plateau (NIBP), is conceptually distinct from noise-free plateaus because it flattens the entire landscape rather than just probabilistic regions. Because the gradient decays exponentially with circuit depth,

an exponential number of measurement shots is required to resolve a training direction, effectively destroying any potential for quantum speedup. This barrier is independent of the parameter initialization strategy, meaning that once a circuit reaches a certain depth-to-qubit ratio, the hardware noise universally erases the functional history of the parameters. (Wang et al., 2021)

Lie Algebraic Unification of Information Loss

To understand the deeper origin of these plateaus, researchers have developed a Lie algebraic theory that unifies different sources of information loss. The trainability of a circuit is inversely proportional to the dimension of its Dynamical Lie Algebra (DLA); more expressive circuits with larger DLAs lead to more concentrated loss functions (Bansal et al., 2026). Furthermore, the concept of g-purity measures the generalized entanglement of a state relative to this algebra. When noise is introduced, it creates "algebraic decoherence," building generalized entanglement or reducing the generalized locality of measurement operators. This framework proves that if a DLA has an exponential dimension, the circuit will inevitably lose its history and become untrainable, regardless of the specific initial state or measurement used. (Ragone et al., 2024)

Restoring Signal Through Engineered Dissipation

The barren plateau phenomenon can be fundamentally viewed as a curse of dimensionality. When a quantum circuit avoids these plateaus, it is often because the computation has been restricted to a polynomially large subspace. While staying in these smaller subspaces makes the circuit trainable, it also potentially makes it classically simulable (Singh et al., 2024)

This creates a "soft dequantization" dilemma: many architectures provably free of barren plateaus can be efficiently surrogated by classical algorithms, provided some initial data is collected from a quantum device. Therefore, the structural traits that allow a circuit to retain its history and training signal are often the same traits that allow a classical computer to emulate the system's behavior, posing a challenge for finding a true quantum advantage. (Cerezo et al., 2025)

Conclusion

The tendency of deep quantum circuits to "forget" their history through noise-induced barren plateaus is a defining challenge for the NISQ era. This information loss is driven by the contractive nature of noise, the specific scaffolding of hardware connectivity, and the exponential dimension of operator algebras. However, by understanding these mechanisms, researchers are developing pathways toward resilience. Whether through the identification of classically simulable subspaces or the use of engineered dissipation to localize training signals, the focus is shifting toward architectures that balance expressiveness with physical trainability. Overcoming this crisis is the key to ensuring that quantum circuits can scale and ultimately deliver on their promise of computational transformation.

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The Role of Memory in Large Language Models: Context Windows, Limitations, and Emerging Extensions

ARTICLE

Abstract

The advent of Large Language Models (LLMs) has led to the creation of advanced models that can produce coherent and context-appropriate text outputs. Despite being intelligent in nature, however, the models lack any form of memory persistence and exist in a limited contextual environment. The paper considers the idea of memory in LLMs by considering the use of context windows in terms of their ability to store information temporarily and limitations, as well as newer concepts like RAG and vector memory systems. It concludes that the development of AI is progressing towards more context-oriented and persistent models despite remaining architecturally limited.

INTRODUCTION

The quick evolution of Large Language Models has resulted in substantial changes for both natural language processing and human-computer communication. The applications using transformer-based architecture exhibit their capacity to produce human-like messages, perform reasoning and even help to implement complex workflow [1], [2]. However, despite all these features, one essential issue still exists—the absence of memory. While traditional software solutions retain the current state between sessions, LLMs lack this feature. They are completely stateless models that treat each entry independently without any context until it is supplied explicitly [2].

Thus, it poses a crucial question of how LLMs work with memory and what the limits of this process are.

In this article, we discuss the methods of dealing with context information within LLMs, as well as their limits and new approaches developed to expand memory [9]–[11].

Memories in Large Language Models

In biological beings, memories refer to the retention and recall of knowledge. However, LLMs utilize contextual representations, wherein all necessary information should fit within the input sequence.

The basic building blocks of this context consist of tokens – pieces of text handled by the machine learning model [4], [7]. All tokens combined constitute the context window, which acts as the temporary working memory for the model.

LLMs lack any form of permanent memory other than the context window. After completing a session or when the context limit has been reached, any information preceding that will be forgotten. Hence, the perception of memory in LLMs is more accurately described as a form of short-term contextual consciousness.

Foundations of Memory in Architectures Used by Transformer Models

Transformer-based Large Language Models (LLMs) make use of a self-attention mechanism as a key building block that enables

storage. Unlike memory-based models, transformers do not store data directly but rather derive relations between tokens while performing inference [1].

The self-attention mechanism allows each token in a sequence to focus on every other token in the same sequence, enabling the analysis of contextual dependencies and semantic relations [1][4]. This constitutes the underlying principle of memory simulation for LLMs as they can only make use of information contained within an input sequence.

Conversely, the self-attention mechanism entails some computational limitations due to quadratic growth in the number of computations performed during attention. This implies that the notion of memory in LLMs is closely associated with efficient and scalable attention mechanisms[6].

Context Window: The Natural Memory System of the Model

The context window determines the largest amount of tokens that can be processed by the LLM at any point in time. The context window serves as the memory space where the model carries out its attention operations [5], [6].

The context window size in contemporary LLMs is becoming larger, with some models having context windows that hold over 100,000 tokens. This development has resulted in:

- The ability to conduct long conversations
- Better document analysis capabilities
- Better coherence when processing long texts

Memory Limitations of LLMs

Restricted Context Window Size

The primary constraint is the fixed window size. Once it exceeds its capacity, it becomes impossible for the model to retain prior information [1].

Degradation of Contextual Attention

With increasing input length, the capacity of the model to pay equal attention to each token decreases. This implies a degradation in remembering the initial part of the input.

No Built-In Persistence

Inherently, large language models cannot remember information between sessions. They have to be reminded in each instance of their interaction.

Heavy Computation

Attention mechanisms are computationally intensive and have quadratic time complexity. Therefore, larger window sizes would require more computation.

Novel Enhancements for Memory Support

RAG Systems (Retrieval-Augmented Generation)

The RAG architecture allows for the retrieval of useful external information prior to responding with a generated message, which means the system has access to large amounts of information to work with, acting as long-term memory, but without expanding the model [3].

Vector Databases and Embeddings

With vector databases, high-dimensional representations (or embeddings) of text data are used. With similarity searches, the LLM systems may retrieve relevant contextual information [7].

External Memory Models

In modern AI systems, external layers for storing user interactions and other historical data are implemented, providing the system with a memory feature that does not rely solely on the model itself [9].

Models with Extended Context Window

The trend among researchers at present revolves around extending the model's context window and making it able to process large amounts of information, despite its inefficiency.

Future Directions

Directions for the Future

The development of memory for large language models seems to be developing into architectures that will allow for internal contextual understanding alongside external memory. Some future trends can be:

- AI memory for the individual user
- Efficient attention models for contextual understanding
- Symbolic and neural memories

Such developments attempt to fill the void between contextual inference and long term knowledge acquisition.

Conclusion

Large Language Models are indeed a major milestone in the development of artificial intelligence technology; however, the absence of inherent memory is an important deficiency to address. With the help of context windows, Large Language Models mimic short-term memory; however, this process still faces limitations due to the limited nature of its storage and non-permanent nature [9]–[11].

RAG and vectorized memory models are becoming increasingly popular in the realm of AI technologies [3]. Moving forward, the implementation of scalable memory architectures can be considered an important element in developing more advanced AI solutions.

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(2023-2027)**

Generative AI and the Need for Human Oversight

ARTICLE

Abstract

Generative AI has moved fast — maybe too fast for the institutions meant to govern it. This article takes an honest look at what happens when powerful language models and image generators run with minimal human checks. We explore where oversight has failed, why it matters beyond regulatory box-ticking, and what organizations can realistically do to keep a human hand on the wheel. The goal isn't to slow AI down; it's to make sure that when it moves, it moves in the right direction.

Keywords

Generative AI, Human Oversight, AI Governance, Responsible AI, Large Language Models, AI Safety, AI Ethics

INTRODUCTION

Think about the last time you used a chatbot to draft an email, looked up a medical symptom and got a surprisingly detailed answer, or watched a video that — only later — you realized wasn't real. These aren't edge cases anymore. Generative AI has quietly embedded itself into the texture of everyday life, and most people haven't fully registered how deep it goes.

That wouldn't be a problem if these systems were reliably correct, fair, and transparent. But they're not — at least not consistently. They confuse facts. They reflect the biases of whoever wrote the internet. They can be nudged into producing content that no reasonable person would sign off on. [5] And yet, in a growing number of workplaces, hospitals, courtrooms, and newsrooms,

their outputs are being acted upon with very little second-guessing.

This article is about that gap — the space between what generative AI can do and what humans are doing to make sure it does it responsibly. Human oversight isn't a bureaucratic add-on. It's arguably the most important design decision in how these systems get deployed. Getting it right matters — not just for regulators and AI developers, but for anyone who interacts with these tools, which at this point is almost everyone.

2. What Generative AI Actually Does — and Doesn't

Generative AI refers to models trained on enormous datasets to produce outputs — text, images, audio, code — that resemble what humans create. The underlying technology, mostly Transformer-based architectures, works by predicting statistically likely continuations of input patterns. [5] That's a simplified description, but it captures something important: these systems are not reasoning in the way humans reason. They're very good at pattern completion.

That distinction matters more than it might seem. A doctor doesn't just recall patterns from medical textbooks; they assess the person in front of them, ask follow-up questions, notice inconsistencies, and apply judgment developed over years. A language model does none of that. It generates text that looks like a knowledgeable doctor might have written it. Sometimes it's accurate. Sometimes it isn't. [1][2]

Where GenAI genuinely excels

- Drafting and summarizing large volumes of text quickly
- Writing and debugging code across dozens of programming languages
- Generating creative content — marketing copy, product descriptions, visual concepts
- Translating languages and making information accessible across literacy levels
- Screening large datasets for patterns that humans would miss or tire of looking for [6][7]

Where it routinely falls short

The failures of generative AI tend to cluster around a few persistent problems. Hallucination — the model confidently stating things that are simply wrong — is probably the most cited. But there's also the subtler issue of bias. Training on internet data means absorbing the full spectrum of human expression, including its prejudices. Models have been caught associating certain professions overwhelmingly with one gender, scoring job applications differently based on names that signal ethnicity, and producing harsher outputs in some languages than others. [3][4]

Then there's the dual-use problem. The same capability that helps a novelist brainstorm plot ideas can help someone craft a convincing phishing email. The same image model that generates product mockups can generate non-consensual imagery of real people. These aren't hypothetical risks. They've happened, repeatedly, and the speed of generative output makes them hard to contain once they start.

3. What Actually Goes Wrong Without Oversight

It's easy to talk about AI risks in the abstract. It's more useful to look at what has actually gone wrong when human checks were absent or inadequate.

In 2023, a New York lawyer submitted a legal brief that cited several court cases — all of them fabricated by ChatGPT. The lawyer hadn't verified the citations. The cases didn't exist. The judge was not amused, and the attorney faced sanctions. This wasn't a story about AI being malicious; it was a story about a professional trusting a tool without understanding its limitations.

Around the same time, a major airline's chatbot gave a passenger incorrect information about bereavement fare policies. The passenger booked a flight based on that information, was denied the discount, and sued. The court ruled in the passenger's favor. The company had deployed the chatbot without building in any mechanism for human escalation on sensitive policy questions.

During the run-up to elections in several countries in 2024, AI-generated audio clips of political candidates saying things they'd never said circulated widely on social media. By the time fact-checkers caught up, the content had already shaped opinions. The speed of generation vastly outpaced the speed of correction.

These examples share a common thread: the technology ran ahead of any human process designed to catch its mistakes. Oversight wasn't absent because nobody thought it was important — it was absent because nobody had built it in, and nobody had thought carefully about where the failure points would be.

4. Why Oversight Isn't Optional

4.1 AI systems optimize for proxies, not outcomes

When you train a model, you give it a measurable target — minimize prediction error, maximize user engagement, score high on a benchmark. But what you actually want is harder to measure: accurate, fair, genuinely helpful outputs. These things are related but not identical. A model can ace every benchmark and still produce outputs that cause harm in the real world, [5] because benchmarks don't capture everything that matters. Human oversight bridges that gap — it's the mechanism by which unmeasured values get enforced.

4.2 Errors compound when no one's checking

In any system where outputs feed into downstream decisions — where a GenAI-drafted document gets approved without review, or where an automated scoring system's results are never audited — mistakes compound. One wrong output influences a decision, which influences a process, which influences another decision. By the time someone notices something's off, the error has been replicated dozens of times. This is why oversight needs to happen early in the pipeline, not just at the end.

4.3 Legal accountability requires a human somewhere

Most legal systems weren't built for autonomous machines. When an AI system causes harm — produces a defamatory statement, denies someone a loan based on a protected characteristic, gives dangerous medical advice — the law needs someone to hold accountable. Right now, that someone is typically

the organization that deployed the system. Human oversight is how that accountability stays real rather than theoretical. Remove it, and you're left with harm that happened, nobody who caused it in any legally meaningful sense, and a victim with no clear recourse.

4.4 Trust has to be earned, not assumed

Public adoption of AI in sensitive domains — healthcare, education, legal services — depends on trust. And trust, as anyone who's studied it knows, is slow to build and fast to lose. Research from the Pew Research Center and Eurobarometer consistently shows that people are significantly more comfortable with AI-assisted decisions when they know a human has reviewed the result. Meaningful oversight isn't just an ethical requirement; it's a precondition for the broader social license that AI needs to operate in high-stakes environments.

5. The Regulatory Picture

Governments have started catching up, though the pace varies considerably. The EU AI Act — the most comprehensive binding framework to date — classifies AI systems by risk level and requires that high-risk applications maintain human oversight, keep audit logs, and allow for human intervention at any point in the decision pipeline. High-risk categories include AI used in hiring, credit scoring, healthcare, and criminal justice.

The US approach has been more sector-by-sector and guidance-based, though the 2023 Executive Order on AI required federal agencies to establish AI governance structures and mandated safety testing for frontier models before deployment. China has introduced content-specific regulations requiring human editorial control over AI-generated content

distributed on platforms.

What's notable across all of these frameworks — despite their differences in scope and enforcement — is that human oversight appears as a non-negotiable baseline. It's not treated as one option among many; it's framed as a prerequisite for responsible deployment. That consensus didn't emerge from nowhere: it reflects a shared recognition that without human checkpoints, accountability breaks down entirely.

6. What Meaningful Oversight Actually Looks Like

Oversight is one of those concepts that sounds self-evident until you try to implement it. What does it mean in practice? What does it cost? And how do you build it into organizations where AI adoption is moving fast and institutional processes tend to move slow?

6.1 Human-in-the-Loop Design

The most direct form of oversight is routing AI outputs through human review before they have consequences. In practice, this works best when it's selective — flagging outputs that fall below a confidence threshold, involve sensitive categories, or affect high-stakes decisions, rather than requiring human review of every single output. A content moderation system, for instance, might automatically handle clear-cut cases but escalate ambiguous ones to a human reviewer. That balance isn't always easy to calibrate, but it's achievable and significantly reduces the risk of unchecked errors at scale.

6.2 Red-Teaming Before Deployment

Red-teaming means deliberately trying to make a model fail before it goes live — probing for biases, attempting to elicit harmful outputs, testing edge cases that might not appear in standard benchmarks. It's borrowed from security practice

, and it works on a similar principle: you want to find the weaknesses yourself before an adversary does.

Major AI labs now conduct structured red-teaming exercises, though the quality and depth varies. For organizations deploying third-party AI tools, red-teaming should include testing in the specific context of use, not just relying on the vendor's general safety evaluations.

6.3 Audit Trails

If an AI-assisted decision turns out to be wrong — or discriminatory, or harmful — you need to be able to trace what happened. That requires logging: keeping records of inputs, model versions, outputs, and the human decisions made at each review stage. This sounds obvious, but many early AI deployments skipped it, either to save storage costs or because nobody had thought through what accountability would look like after the fact. Audit trails aren't just for legal cover; they're also how organizations learn from failures and improve their processes over time.

6.4 Building Oversight Into Culture, Not Just Process

The hardest part of oversight isn't the technical infrastructure — it's the human behavior. People under time pressure will skip review steps. Workers who trust a tool's track record will stop checking its outputs. Managers who've never seen an AI failure will deprioritize oversight budgets. Building a culture where oversight is taken seriously — where people understand why it exists and what can go wrong without it — is slower work than deploying a monitoring tool, but it's ultimately what determines whether formal oversight mechanisms actually function.

Conclusion

The argument for human oversight of generative AI isn't that the technology is bad or that progress should slow down. It's simpler than that: powerful tools require responsible handling, and the more consequential the tool, the more important it is that humans stay engaged with how it's being used.

Generative AI is now consequential enough that leaving it unsupervised — even partially, even temporarily — creates real risks for real people. The lawyer who submitted fake citations, the airline passengers given wrong information, the voters exposed to synthetic disinformation: none of them were victims of hypothetical risks. They were victims of actual deployments where oversight was underbuilt or absent.

The good news is that oversight is not technically out of reach. Human-in-the-loop systems work. Red-teaming finds problems before users do. Audit trails create accountability. What's needed isn't invention — it's commitment: from developers, from deploying organizations, and from the policymakers who set the rules of the road. Generative AI will keep advancing regardless. The question is whether the human infrastructure around it advances at the same pace.

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FUN TECH FACTS

1. **Cosmic Origins of Wi-Fi**

In the 1990s, Australian scientists wrote a complex algorithm to untangle messy radio waves from collapsing black holes. They never found the black holes, but that exact mathematical formula became the foundation for modern Wi-Fi!

2. **The One-Ton Hard Drive**

IBM's 1956 "350 RAMAC" was the size of two refrigerators and weighed over 2,000 pounds. Its capacity? A whopping 3.75 Megabytes barely enough for a single MP3! Furthermore, it wasn't for sale; renting it cost an astronomical \$3,200 per month.

3. **Apollo's Hand-Woven Code**

The Apollo 11 computer operated on a mere 4KB of RAM. Its software was stored in "core rope memory," where skilled textile workers physically wove the binary code with copper wire. Passing a wire through a magnetic ring meant a 1, and bypassing it meant a 0.

4. **Proving Humanity, Digitizing History**

Remember those early, wavy reCAPTCHA words you used to decipher? One word proved you were human, but the other was a scanned snippet from an old text that optical character recognition (OCR) couldn't read. Millions of users unknowingly crowd-sourced the digitization of historical archives!

5. **The Wild West of Free Domains**

Before the commercial web exploded, anyone could claim prime internet real estate for absolutely \$0. Domain registration remained completely free until 1995, when the first fee was introduced: a steep \$100 for a two-year lease.

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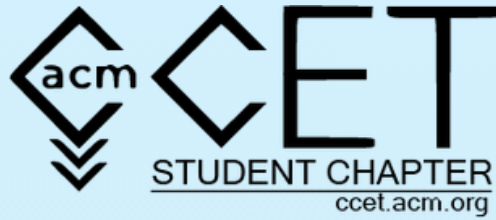
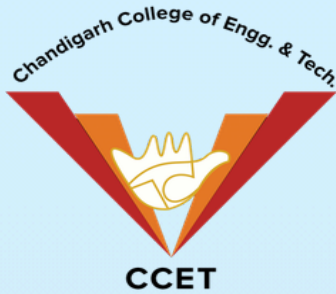
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— Alan Kay

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